

Forecasting LoRaWAN RSSI using weather parameters: A comparative study of ARIMA, artificial intelligence and hybrid approaches

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ABSTRACT

LoRaWAN technology's reliability is challenged by weather parameters, which can influence the communication channel design, especially when dealing with outdoor devices. We propose to analyze this effect by evaluating the relationship between the received signal strength indicator (RSSI) and different weather parameters, as well as its temporal changes. A rigorous statistical analysis of the RSSI sequences is conducted to assess if they could be represented by a specific statistical model. For this purpose, several models are investigated. The Artificial Intelligence (AI) algorithms cover machine learning (ML) and deep learning methods are appealing when dealing with time series forecasting. Nevertheless, the classical autoregressive integrated moving average (ARIMA) model can be an attractive alternative due to its simplicity. Therefore, this work proposes a comparative study of ARIMA, AI, and hybrid approaches to forecast the RSSI using weather parameters as regressors. The considered AI algorithms are the artificial neural network (ANN), support vector machine (SVM), random forest (RF), and Long Short-Term Memory (LSTM). Also, hybrid models are constructed, coupling the ARIMA with them. The models are evaluated in time series of RSSI, measured by eight different LoRaWAN transmitter nodes and considering the temperature, pressure, relative humidity, and rain as weather parameters. Our analysis reveals that temperature is the dominant factor among weather parameters, and negatively affects RSSI. The ARIMA model that uses only the temperature as a regressor provides consistently better fits than the ARIMA without regressors. Moreover, coupling the ARIMA with the temperature as a regressor and the ANN (ARIMA-ANN) is the best option among the pure AI and hybrid approaches. However, it provided accuracy measures very close to those obtained from the ARIMA model fitted in the first stage, with similar performance. Therefore, the ARIMA model considering the temperature is the most competitive alternative when analyzing RSSI measurements, with the advantage of being the most straightforward method. These results suggest that the RSSI from the analyzed LoRaWAN receiver nodes may not present nonlinear patterns and, considering several weather parameters, they are affected mainly by the outdoor temperature.

1. Introduction

The Internet of Things (IoT) consists of a complex ecosystem network of interconnected devices that exchange information over wired or wireless connections, enabling physical things to interact and share crucial information while making choices and carrying out crucial tasks [1]. The IoT has experienced rapid development and its growth is expected to continue since different domains are searching for IoT solutions to enhance their operations [2]. It includes different technologies for connecting various devices, and one of the reasons for its success is the ability of wireless sensor networks (WSNs) to provide

ubiquitous data [3]. Therefore, it can be useful for multiple applications, such as agriculture [4], environmental monitoring [5], road traffic monitoring [6], and smart cities [7].

Among the wireless network technologies supporting IoT, the Long Range (LoRa) and the associated Long Range Wide Area Network (LoRaWAN) have been gaining attention [8]. The advantages of using such technologies are their power efficiency and ability to cover long distances [9]. However, one of the challenges facing LoRaWAN is the impact of environmental conditions in the reliability of radio transmissions and links between nodes [10]. Variations in weather

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parameters can also affect synchronization and the operating time of the network [11].

1.1. Research motivations and applications

Despite the relevance of this topic, the LoRa wireless community lacks publicly available datasets that would allow to associate channel performance measurements and weather parameters [12]. The work [12] addresses this gap by collecting the received signal strength indicator (RSSI) of eight LoRaWAN devices recorded alongside the data from a weather station. The authors give some insights into the correlation between the RSSI and weather parameters. However, further investigation has to be carried out to find relationships between them [12].

On one hand, the utilization of RSSI can be leveraged to derive meteorological parameters from devices lacking dedicated sensors. In this context, RSSI data obtained from a multitude of simple and cost-effective devices can efficiently provide insights into weather conditions, eliminating the necessity for deploying IoT devices equipped with specialized, and hence more costly, sensors.

On the other hand, the ability to determine the location of the nodes is a fundamental aspect of many modern IoT applications. While Global Navigation Satellite Systems can be employed to achieve this goal, more economical and uncomplicated methods are preferable for low-power networks. Consequently, utilizing simple received signal strength indicator (RSSI)-based ranging techniques proves to be an appealing choice for energy-efficient systems like LoRa. Nevertheless, predicting the interactions between RSSI and the real-world environment is challenging, often resulting in significant errors during the localization process. Hence, investigating the influence of weather conditions on RSSI can enhance the accuracy of localization [8].

1.2. Main contributions

In this paper, we examine the role of weather parameters in accurately forecasting the RSSI measurements leveraging on the dataset [12]. We also aim at identifying which weather parameters are more relevant to forecast RSSI, as well as the direction of their impact. These objectives may be pursued through time series models that take weather parameters as regressors. Covering machine learning (ML) and deep learning methods, the artificial intelligence (AI) algorithms are among the state of the art (SoA) approaches. Nevertheless, classical tools have benefits for addressing forecasting problems. In cases where several methods give roughly equivalent performance we can apply the Occam's razor principle and pick the simplest [13]. Therefore, the classical autoregressive integrated moving average [14] (ARIMA) model can be an attractive alternative because it can generate statistically significant solutions in linear data and hold strong interpretability and simplicity [15].

These models have proven essential for a wide variety of forecasting applications, such as inflation rates [16], vehicle sensor data [15], and detection problems [17,18]. Despite their success, they have been applied only limited to LoRaWAN technology. In particular, [19] employed the ARIMA models to forecast the temperature and humidity collected from a LoRaWAN-based sensor with the purpose to present a scalable smart air quality monitoring system. Moreover, these models were considered in [20] as a ML technique to forecast soil humidity by using the RSSI. On the contrary, our work focuses on forecasting RSSI measurements using weather parameters.

While ARIMA models excel in providing solutions for linear data, their effectiveness diminishes when faced with nonlinear patterns. Indeed, they primarily assume linearity and stationarity in the underlying time series. In these cases the AI techniques are useful, such as the artificial neural network (ANN), support vector machine (SVM) and random forest (RF). The Long Short-Term Memory (LSTM) is a deep-learning architecture designed to capture long-range dependencies. It is frequently applied to challenges dependent on time series data.

Finally, in time series analysis, hybrid approaches have been developed to improve forecasting accuracy [21]. The hybridization can be performed using series models, which decompose the data into two main components and analyze them separately [15]. In this context, [22] pioneered a hybrid approach by combining the ARIMA model with the ANN algorithm. Besides ANN, other ML algorithms can be explored in hybrid approaches. Indeed, ARIMA have been combined with SVM and RF algorithms [23,24]. Despite hybrid approaches are increasingly gaining attention due to their potential to lead to more accurate predictions [25], to the best of authors' knowledge, they have not yet been employed for predicting channel performance in LoRaWAN technology.

Hence, the novelty of the present work consists of a rigorous statistical analysis of the RSSI sequences to assess if they could be represented by a specific statistical model. For this purpose, a comparative study is conducted for the first time among ARIMA, AI, and hybrid approaches to forecast the RSSI using weather parameters as features. The main contributions of this paper are the following:

- A statistical representation of RSSI sequences is conducted, where both time dependency and weather parameters are included.
- Using an appropriate statistical model, we identify the subset of features that most affect the RSSI values, extracting the temperature as a relevant biophysical parameter.
- All R codes and implementations developed are available to the public on the GitHub platform.¹ Sharing the computational codes facilitates the understanding of the results and encourages advancements in the methods applied, contributing to the replicability of our research. The LoRa wireless community lacks publicly available datasets [12], as well as the related computational codes.

The rest of this work is structured as follows: Section 2 outlines the background and related works. Section 3 presents the proposed methodology and briefly describes the dataset under investigation [12]. Section 4 is devoted to the RSSI estimation and analyzes the contribution of the weather parameters in forecasting accuracy. Finally, Section 5 presents the concluding remarks.

2. Background and related works

2.1. Background on lora technology

The LoRa communication system utilizes data transmission for communication and localization purposes, offering the advantage of saving power and enabling indoor tracking without relying on an external GPS device [8]. The LoRa technology is designed for low-data rate wireless communication, making it well-suited for IoT applications such as reliable bi-directional communication, mobility, and localization services [10]. While LoRa defines the physical layer features, LoRaWAN specifies the system architecture and network protocols by providing a medium access control (MAC) mechanism that facilitates communication between multiple LoRa end-devices and a LoRa gateway using LoRa modulation [26]. The LoRa physical layer utilizes chirp spread spectrum (CSS) modulation, which encodes information through chirps with a linear frequency variation over time. The key parameters in LoRa modulation are bandwidth (BW), spreading factor (SF), and code rate (CR), which determine the bit rate, interference robustness, and decoding ease [26]. Using f_0 as the center frequency of the sweep interval $[f_0 \pm \frac{1}{2}BW]$ and $t = 0$ as the signal beginning moment, the LoRa chirp signal can be represented mathematically as [11]:

$$s(t) = A_0 \cos(2\pi t + 2\pi \int_0^t \Delta f(m, t) dt + \phi_0), \quad (1)$$

¹ <https://github.com/renata-rojasg/LoRaWAN/>

where $0 \leq t \leq T_m$, $A_0 > 0$ represents the amplitude of a chirp, $m \in \{0, \dots, M-1\}$ denotes the modulation symbol, $\Delta f(m, t)$ is an instantaneous frequency offset dependent on symbols, ϕ_0 describes the signal phase at $t = 0$, and T_m represents the chirp duration. The symbol duration is related with the BW and SF by the equation [8]:

$$T_m = \frac{2^{\text{SF}}}{\text{BW}}, \quad (2)$$

where $\text{SF} = \log_2(n_{\text{chirps}})$, n_{chirps} is the number of chirps per symbol and a symbol contains 2^{SF} chirps, which covers the whole band [8].

The LoRa chirp signal is closely tied to channel characterization, primarily through the measurement of RSSI. The RSSI quantifies the received signal's power level and offers valuable insights into channel conditions. The RSSI values are usually represented as [8]:

$$\text{RSSI} = - (10n \log_{10} d - A) \quad (3)$$

where n represents the loss parameter of the environment, d is the distance between transmitter and receiver, and A indicates the received power at a reference distance of 1 meter and is measured in dBm. The parameter A is determined by the physical characteristics of the radio device, whereas the value of n is strongly influenced by the environment and the operating frequency [27]. Therefore, variations in RSSI provide valuable information about channel characteristics, enabling the optimization of system parameters, and evaluation of link quality.

2.2. Related works

The reliability and connectivity of wireless channel transmissions can be affected by fluctuations in weather parameters regardless of the employed modulation [28]. This effect is especially relevant for outdoor WSNs [29,30] since they are more exposed to environmental changes such as precipitation, temperature, and humidity. A summary of research works that related RSSI with temperature, humidity, rain, and fog was presented in [30]. Most of these works revealed a negative correlation between temperature and signal strength and a negative relation with humidity, which are in line with the results of this work. However, some contradictory results were found regarding the dominating factor and the direction of the relationship [30], and about the relation between RSSI and temperature, humidity, and rainfall [31]. The reasons may be due to methodological differences such as meteorological data sources, experiment duration, and communication hardware, and the peculiarities of each experimental design should be considered [31].

In this context, the impact of weather parameters for LoRaWAN technology has been widely discussed in the recent literature, and here summarized in Table 1. Temperature and humidity were the most investigated weather parameters, and most of the works revealed a negative relation between RSSI and temperature, as well as between RSSI and humidity. The work [32] investigated the deployment of LoRaWAN under different temperature conditions in Northern Sweden. Measurements were performed during eight months with temperatures ranging from -28.7 to 24.8 °C. By comparing cumulative distribution functions (CDFs), the authors concluded that signal strengths benefited from the cold. Similar results were reported by [10] for an experiment designed in Bosnia and Herzegovina. It was verified that higher temperature and relative humidity led to the degradation of network performances. A negative relation between RSSI and temperature was also verified in [33]. Specifically, temperature, humidity, atmospheric pressure, and CO₂ measures were used to develop a protocol for increasing the efficiency and predicting the LoRaWAN communication effectiveness. The experiment was deployed in Ireland and corroborated the use of environmental data for these purposes. The conclusion was that it was possible to reduce the transmitted power when considering information from environmental sensors. Additionally, [34] computed the temperature, humidity, atmospheric pressure, and rain and illustrated

their relationship with signal strength by using graphical tools and descriptive statistics. It was proven that RSSI and SNR performance worsened in rainy weather and when humidity increased.

On the other hand, a few works reported different findings [28,35]. In particular, a positive correlation was found between RSSI and the atmospheric temperature [35]. Since this pattern was not typical in the literature, the authors observed that it could be attributed to the prevailing meteorological conditions in Malaysia [35]. To the best of our knowledge there are no other works carried out in tropical regions. They also stated that the RSSI values are not correlated with humidity and rainfall rates. An analysis of RSSI under temperature and humidity was also presented in [28], which found a discrepancy only for the humidity, while it was in line with the SoA for the temperature. The results were based on transmission tests sorted out by placing a LoRaWAN transmitter within a climatic chamber. The study corroborated that such outcomes affected the LoRaWAN transmission performance. A performance drop was noted as the temperature increased. However, nonlinear behaviors were displayed from the graphical representations, indicating that RSSI and temperature might be adequately approached by a fifth-order polynomial curve rather than by a linear trend. Regarding humidity, it was observed that increased water content in the air improved communication capabilities [28].

Finally, a study was conducted in [11] that considered temperature, humidity, pressure, and rain to predict LoRaWAN performance indicators. To achieve this purpose, the authors used the Bayesian surrogate Gaussian process-based bidirectional Long Short Term Model stacked autoencoder (BSGP-BLSTM-SAE). They also computed Pearson's correlation, which verified that all weather parameters had a statistically significant correlation with the performance, except the rain. They observed a negative correlation between RSSI and temperature, aligning with the majority of existing studies. However, a disparity was noted for the relation between RSSI and humidity, revealing a positive association. The correlation was higher for RSSI than for SNR in most cases.

In summary, it can be seen from Table 1 that all the studies considered evaluate the temperature effect and concluded that it significantly impacts the performance indicators. It is generally understood that RSSI and temperature show a negative relation since higher values of temperature can increase thermal noise, which can worsen signal strength. That is the conclusion of most related works, with the exception of [34, 35]. The authors in [34] argued that such behavior can result from the superimposition of other outdoor effects that have a more prominent extent than the temperature in the area under investigation, the city of Piombino, Italy. Indeed, during the data collection period, the authors verified a wide range of different relative humidity (from 10% to 100%) and atmospheric pressure (from 998 hPa to 1026 hPa) [34]. Also [35] reported a positive relationship between the RSSI and the temperature. It is also worth noting that their experiment was performed in a tropical region, where the average temperatures are higher than in other related works. Indeed, while the temperature range was between 25° and 45° Celsius in [35], it was between 0° and 17° Celsius in [33] and between -5 and 20° Celsius in [11,12].

As shown in Table 1, humidity is the second most frequently analyzed weather condition. It appears in all the works except [32], which focused only on the temperature. Most of the results show this variable has a negative effect on the signal strengths. This is foreseeable since the higher humidity results in denser water vapor in the air, which may degrade the signal performance [34]. However, a few works reported a positive relationship of the humidity with the SNR [33], or with the RSSI [28], or for both SNR and RSSI measures [11], which may be due to the superimposition of other weather parameters. Indeed, the temperature may influence the relative humidity, which may affect the correlation results [33].

Finally, few effort has been spent for pressure and rain, which appeared only in three studies [11,34,35]. In the case of rain, its effects are significant only in [34]. This may be related to the range

Table 1

Comparison of findings related to weather parameters effects and signal strengths computed from LoRaWAN technology.

Reference	Country	Method	Metric	Relation with weather condition				
				Temperature	Humidity	Pressure	Rain	Note
[32]	Sweden	CDFs	RSSI SNR	negative negative				Cold weather improves the signal strengths.
[10]	Bosnia and Herzegovina	Descriptive statistics graphics	RSSI SNR	negative negative	negative negative			Increase of temperature and humidity leads to performance degradation.
[33]	Ireland	Pearson's correlation decision trees	RSSI SNR	negative negative	negative positive	negative positive		The weather parameters are useful to predict the effectiveness of communication.
[34]	Italy	Descriptive statistics graphics	RSSI SNR	slightly positive slightly positive	negative negative	none none	negative negative	Superimposition of other outdoor effects may have a more extent than the temperature.
[35]	Malaysia	Pearson's correlation	RSSI	positive	none		none	Increase in solar radiation also increases the onboard and atmospheric temperature. Experiment performed in a tropical region.
[28]	Simulation experiment	Descriptive statistics graphics	RSSI	negative	positive			Performance drop as the temperature increases. The humidity proportionally improved the performances.
[11]	Italy	Pearson's correlation BSGP-BLSTM-SAE	RSSI SNR	negative negative	positive positive	negative negative	none none	The magnitude of the impact was smaller for the SNR than the RSSI for most cases.

considered for these weather parameters. Indeed, [34] experienced a wide interval from clear sky to heavy rain and fog, while [11] mainly observed dry weather during the experiment, hence the relation between RSSI and rain could have been less evident.

Although most of the SoA confirm a negative relation of RSSI or SNR with temperature and humidity, further investigation is still required to consider the impact of all weather parameters to LoRaWAN technology. Effects to be considered during the evaluation include the duration of the experiments, since the direction and intensity of the relations may be subject to seasonal effects, as well as the mutual interaction among the weather parameters [35]. Furthermore, it is worth to note that most of the methodologies from Table 1, such as descriptive analysis and Pearson's correlation, did not consider the changes over time. However, if the signal strength presents a serial dependence over time, it is necessary to consider models that adequately accommodate such characteristic, as proposed in the following.

3. Proposed methodology

The dataset that we analyzed [12] comprises RSSI measurements from a testbed consisting of one LoRaWAN gateway and eight LoRaWAN nodes deployed in a vineyard area in Italy. Weather data is also collected, including the temperature (T), barometric pressure (Bar), relative humidity (RH), and rain. All the variables are provided with hourly frequency, which may require time series models for accurate analysis and forecasting.

In this context, ARIMA models are popular time series models able to capture the temporal dependencies and linear patterns present in the data. Moreover, it is possible to consider ARIMA models with regressors to evaluate the effects of factors, providing a more comprehensive understanding of the dynamics of the target variable and its relationship with explanatory variables. Finally, combining the ARIMA and ML techniques through hybrid approaches offers a powerful approach to addressing nonlinear dynamics that may be present in the data.

Therefore, we propose conducting an extensive study to compare the performance of ARIMA models and hybrid approaches for forecasting RSSI under varying weather parameters employed as regressors. In the rest of this section, we discuss the mathematical formulation of the model employed in the study and provide a detailed description of the analyzed dataset.

3.1. ARIMA model

The signal strength forecast is computed using the ARIMA approach with and without the presence of regressors. The ARIMA model is a combination of autoregressive (AR) and moving average (MA) components with the integration filter (I), enabling the modeling of the series of the three components or a subset of them. Let $y_1, \dots, y_n$ be an RSSI time series of interest. The ARIMA(p, d, q) takes the following form

$$\nabla^d y_t = (1 - B)^d y_t, \quad (4)$$

where

$$y_t = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j r_{t-j}, \quad (5)$$

∇^d refers to the difference operator of order d , B is the backward shift operator, $\alpha \in \mathbb{R}$ is a constant, $\phi_1, \dots, \phi_p$ are the coefficients of the autoregressive terms with order p , $\theta_1, \dots, \theta_q$ are the moving average coefficients with order q , and r_t is the white noise series that assumes Gaussian distribution with zero-mean and constant variance.

The methodological procedure of the ARIMA modeling process is called Box-Jenkins methodology [14]. It consists of three main steps:

- Identification: determining the AR, I, and MA orders of the process that generates the series, i.e., the number p and q of the coefficients ϕ_i and θ_j . In this step we also check the series' stationarity and transforming it if necessary, and determining the model through the analysis of Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF);
- Parameter estimation: once you have the filters representing the time series, the next step is to estimate the parameters of the ARIMA model. This involves obtaining estimates of α , $\phi_1, \dots, \phi_p$, $\theta_1, \dots, \theta_q$, i.e., finding the coefficients of the AR and MA terms. Various methods can be used for parameter estimation, and in this paper, we use the maximum likelihood method.
- Diagnostic checking: determine whether the adjusted model can describe the behavior of the values of the RSSI time series being analyzed.

After selecting the model that adequately fits the RSSI sample time series data, we perform the time series prediction and evaluate its accuracy. The most commonly used accuracy measurements to identify the best fit among the competing models are the Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root-Mean-Square Error (RMSE). The smaller their values, the more accurate the model.

In this work, we use the forecast package implemented in the R programming language to fit the ARIMA models. Moreover, we fit a regression model with ARIMA errors in this package to consider the effect of regressors in the analysis. Denoted as ARIMAX, the regression model with ARIMA errors is given by

$$y_t = \sum_{l=1}^k \beta_l x_{lt} + \epsilon_t \quad (6)$$

where $\epsilon_t = \nabla^d \epsilon_t$ is an ARIMA process defined analogously to (4), $x_{1t}, \dots, x_{kt}$ are the explanatory variables, and $\beta_1, \dots, \beta_k$ are their corresponding coefficients (β -coefficients).

From this model, y_t represents the series of RSSI and can consider the weather parameters as regressors. Therefore, we can compare different configurations for the ARIMAX and verify which weather parameters can improve the RSSI forecasting results, and the direction of the relationship.

3.2. AI algorithms

The general framework of AI refers to the simulation of human intelligence by a system or a machine [36]. This research field encompasses ML and deep learning, particularly suited for addressing highly complex problems characterized by nonlinear behaviors. Then, by training models, AI frameworks can learn the internal laws of data for supporting and realizing an extensive range of applications [36]. ANN, SVM and RF methods are some of popular ML algorithms [37].

The architecture of ANN consists of input, hidden (middle), and output layers to model complex relationships between sets of input and output variables [38]. Comprising interconnected nodes, or artificial neurons, organized into layers, ANNs process information through weighted connections to mimic how biological neurons transmit signals. The network is trained using a learning algorithm to adjust these weights, allowing it to recognize patterns, make predictions, and perform complex tasks. ANN is used in time series forecasting because it is a standard solution with a high accuracy rate when the nonlinearity in the input data is identified [39]. The ANN can be fitted using the `neuralnet` function in R programming language [40], which is performed using as default one hidden neuron in each layer and the logistic activation function.

The SVM approach is based on support vectors and a kernel function. First developed by [41], it is widely used to deal with classification problems to accommodate nonlinear class boundaries [13], but they have also been applied successfully for regression tasks [42]. The method is usually referred to as support vector regression when used for regression. By transforming the input data through a nonlinear mapping into a high-dimensional feature space and executing a linear regression within this space, it becomes possible to identify a nonlinear regression line within the original input space [42]. The SVM can be fitted using the `svm` function in R programming language [43], which is performed using the radial kernel as default.

Finally, the RF consists of a collection of several randomized regression trees [44]. During the training process, each tree in the forest is performed using a random subset of the features, thereby decorrelating the trees [13]. The final prediction is often an aggregation of predictions from individual trees. An interesting aspect of the RF algorithm lies in its ability rank the importance of the input variables over the pattern variability of the target variable [37]. The RF can be fitted using the `randomForest` function in R programming language [45], which is performed using one third of the variables randomly sampled as candidates at each split and 500 trees to grow.

The referred ML algorithms are conventional supervised learning approaches frequently used for filtering and predicting time series issues, see [37,39,42] for instance. While these models can be fitted using lagged outcomes as features, they do not inherently incorporate recurrent dependence in their construction. Therefore, one feasible alternative is the utilization of deep learning techniques such as the

LSTM network. The LSTM [46] is one type of recurrent neural network designed to capture and model sequential dependencies in data. It makes them particularly well-suited for time series analysis, where the temporal aspect and recurrence play a crucial role [13]. The LSTM network has recently been used to solve networking problems. Authors of [47] use LSTM to study real-time mobile bandwidth prediction in various mobile networking scenarios. A recent work on [48] use LSTM to incorporate wastewater-based epidemiology information to nowcast and forecast new weekly COVID-19 cases. In this work, we can fit the AI algorithms considering the series of RSSI as outcome and its lagged values as features plus the weather parameters. Therefore, we can also use the weather parameters to forecast the RSSI measurements. In recent years, many researchers have proposed a wide range of AI approaches using Wi-Fi RSSI fingerprinting schemes.

3.3. Hybrid methodology

Since the ARIMA model alone is insufficient for nonlinear data management [22], besides testing pure AI algorithms, we develop a hybrid framework coupling the ARIMA model and AI techniques. Hence, we assume that each RSSI time series L_t is described by linear and nonlinear components, according to

$$L_t = Y_t + S_t, \quad (7)$$

where Y_t denotes the linear component, and S_t is the nonlinear part. The two components of the time series are estimated using a two-stage framework, where the linear part is obtained from the ARIMA model in the first stage and its residual is represented as

$$r_t = L_t - \hat{Y}_t, \quad (8)$$

where $\hat{Y}_t$ denotes the linear output of the ARIMA model.

In the second stage, an AI algorithm is employed to the residuals in Eq. (8), which were computed in the first stage. Thus, we represent the residual of the ARIMA model as follows

$$r_t = \hat{S}_t + \epsilon_t, \quad (9)$$

where $\hat{S}_t = f(r_{t-1}, r_{t-2}, \dots, r_{t-\rho})$, $f(\cdot)$ is a nonlinear function determined by the considered AI algorithm, ρ is the number of lagged residuals considered as inputs, and ϵ_t indicates the corresponding random error. Finally, the two-step combined prediction is defined as $\hat{L}_t = \hat{Y}_t + \hat{S}_t$.

Fig. 1 represents the general framework of the considered hybridization. This approach is usually employed to improve the forecasting accuracy and dealing with chaotic features and irregularities in time series data and have been extensively leveraged in various forecasting applications [15]. Here, we consider either the ANN [49], RF [44], SVM [50], or LSTM [46] as AI algorithms in the second stage. The resulting models comprise the ARIMA-ANN, ARIMA-RF, ARIMA-SVM, and ARIMA-LSTM hybrid models, respectively.

3.4. Experimental data collection

The data used in this study are taken from the available open-access literature reported by [12] and analyzed by [11]. They include time series of RSSI measured by eight LoRaWAN transmitter nodes (Tinovi PM-IO-5-SM), namely RSSI_01, ..., RSSI_08. The nodes are LoRaWAN class A devices powered by 3.7v 18650 batteries, they were configured with ABP activation and set to transmit data every 300 s. The gateway, located on a building roof at a height of 12 m, operates in the 863–878 MHz frequency range with a 6.5 dBi omni-directional antenna, and is connected to a public LoRa network server. The weather data are collected from a weather station model Davis Vantage Pro2 and consist of T in °C, Bar in hPa, RH in %, and rain in mm. The combined hourly dataset contains 2029 records with average hourly values collected at a vineyard in Rio Saliceto, Reggio Emilia, Italy. It comprises the period from November 10, 2020, 23:20:00 to March

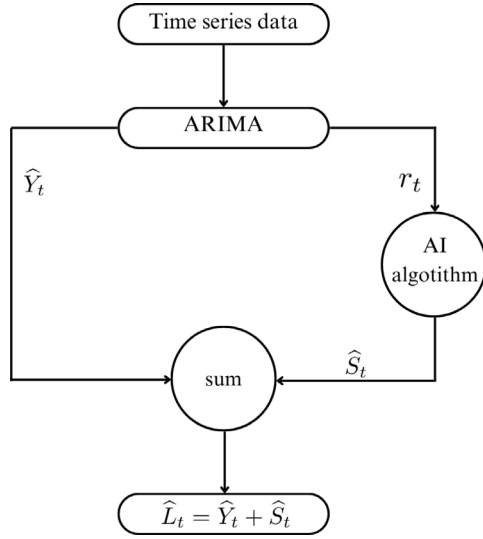


Fig. 1. General framework to forecast time series data combining ARIMA and ML techniques.

Table 2
Descriptive statistics of the studied variables.

Variable	Min.	Median	Mean	Max.	SD	NA
T	-4.27	4.26	4.27	15.35	3.20	0
RH	37.17	81.00	78.90	86.83	7.15	0
Bar	985.65	1015.48	1014.60	1035.15	10.30	0
Rain	0.00	0.00	0.170	15.50	0.74	0
RSSI_01	-58.08	-56.08	-55.80	-51.92	1.12	0
RSSI_02	-92.67	-80.55	-80.40	-71.25	3.46	0
RSSI_03	-70.17	-67.33	-67.26	-63.67	1.00	1
RSSI_04	-84.42	-77.17	-77.25	-71.00	1.86	1
RSSI_05	-81.75	-70.42	-71.10	-62.00	3.51	1
RSSI_06	-80.75	-62.75	-63.23	-53.55	4.61	1
RSSI_07	-67.36	-61.82	-61.82	-58.27	0.88	0
RSSI_08	-91.42	-82.17	-82.66	-69.62	3.32	0

5, 2021, 05:00:00 GMT. However, the environmental variables have eleven missing data corresponding to the period from February 2, 2021, 12:00:00 GMT, to February 2, 2021, 22:00:00 GMT. Thus, we perform the analysis considering the data up to February 2, 2021, 11:00:00 GMT, resulting in a final dataset with 1870 records. The exception is the RSSI_08, which has lower data counts and starts on November 22, 2020, at 15:20:00.

Table 2 provides a descriptive analysis containing the minimum (Min.), median, mean, maximum (Max.), standard deviation (SD), and missing values (NA) of the variables in the final dataset. Five transmitter nodes presented one missing value for the RSSI. We replace them with the mean of the corresponding node to perform the time series analysis and calculate the descriptive measures. Notice there was a low level of rain, with no occurrence in 50% of the days (median = 0.00). The temperatures are also low since the experiments were conducted in the fall and winter. Regarding signal strength, RSSI_01 has the highest values, while RSSI_02 and RSSI_08 present the smallest ones. This behavior can also be observed in Fig. 2, which shows the RSSI values for all the transmitter nodes during the analyzed period. These are the time series we are interested in forecasting. We also compute the Pearson's correlation between each weather condition and the signal strengths.

4. Results and discussion

In this section, we correlate the RSSI with the weather parameters and provide a comparative study of ARIMA and hybrid approaches to forecasting the RSSI using the weather parameters. The correlations

Table 3

Pearson correlation for the weather parameters and signal strengths. Values followed by * have p -value < 0.05.

	T	RH	Bar	Rain
T	1.000			
RH	-0.126*	1.000		
Bar	0.145*	0.054*	1.000	
Rain	0.008	0.000	-0.244*	1.000
RSSI_01	-0.460*	0.146*	-0.144*	0.056*
RSSI_02	-0.225*	0.115*	-0.058*	0.225*
RSSI_03	-0.371*	-0.095*	0.147*	-0.075*
RSSI_04	-0.386*	0.168*	-0.411*	0.285*
RSSI_05	0.189*	0.150*	0.166*	0.022
RSSI_06	-0.479*	-0.263*	-0.090*	0.013
RSSI_07	-0.223*	-0.121*	-0.179*	-0.092*
RSSI_08	-0.043	0.270*	0.401*	0.045

are computed using all the observations in the dataset. For forecasting purposes, each dataset is divided into an 80% ($n = 1496$) training and a 20% ($n = 374$) testing sets to assess the forecasting performance. The exception is RSSI_08, which has $n = 1346$ elements in the training set. The training set is used exclusively for model development, and then the test sample is used to evaluate the established model in one-step-ahead forecasts. In addition to the MAE, MAPE, and RMSE accuracy measurements, we compute correlation (COR) between the actual and forecasted values to analyze the performance of the fitted models in the testing sets. We also provide experiments regarding the usefulness of the fitted models to replace missing values in the dataset.

4.1. Correlating the RSSI and the weather parameters

Preliminary insights are drawn by computing Pearson correlation for each pair of weather parameter and signal strengths time series and testing their significance. The results are presented in Table 3. Most of the correlations are significant at the 5% significance level, the exceptions are the RSSI_05, RSSI_06, and RSSI_08 with the rain, and the RSSI_08 with T. We also note the correlations with T are negative for all nodes except RSSI_05. For the other weather parameters, there is no standard behavior in the direction of the relationships. The correlation is positive for some nodes, while for others it is negative. It may happen because the Pearson correlation measures linear relationships and does not consider the effect of time, which would explain different behavior. For example, using a mean trend function [28] observed that the relationship between the temperature and the RSSI may not be regulated by a linear trend. Indeed, it is important to consider methodologies that include time in the developed model, as well as nonlinearity, as for the ARIMA and hybrid models.

4.2. Fitting the ARIMA model

We fit the ARIMAX complete model considering all the weather parameters as regressors and define the p , d , and q orders using the Box-Jenkins methodology and the forecast package in R programming language [51]. Hereafter, we refer to this complete model as ARIMAX_{All}, while other subscripts are used when only some of the regressors are considered. Table 4 summarizes the p , d , and q orders and provides information about the significance of the regressors. Notice that $p = 2$, $d = 1$, and $q = 3$ is the final order for four nodes, being the most frequent configuration. All the series are differentiated once, i.e., $d = 1$, and include both AR and MA terms. Fig. 3 displays boxplots to illustrate the magnitude of the β -coefficients estimated for each weather parameter in the eight complete final models.

We can also verify that all the weather variables were significant in at least two nodes. The T has a significant coefficient in 100% of the final models, being the dominant factor among the analyzed weather parameters, and having the highest coefficients in terms of absolute values (see Fig. 3). In contrast with the correlation results, which

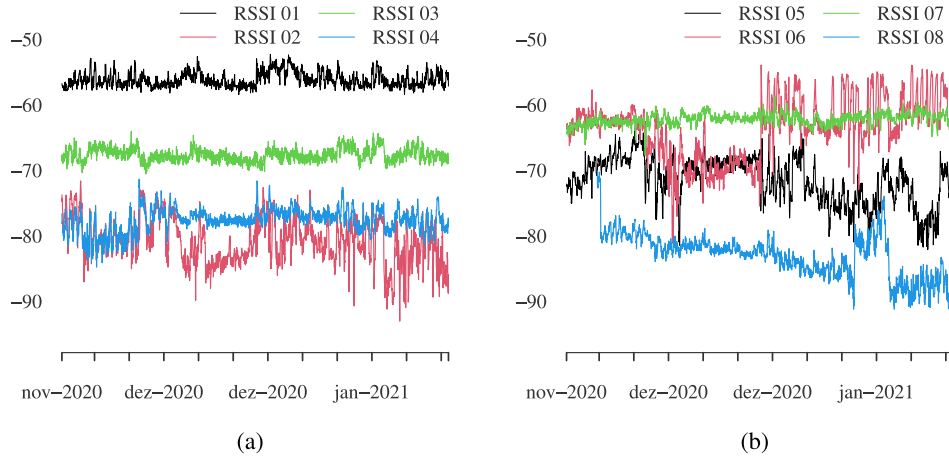


Fig. 2. RSSI measurements hourly average values for each node.

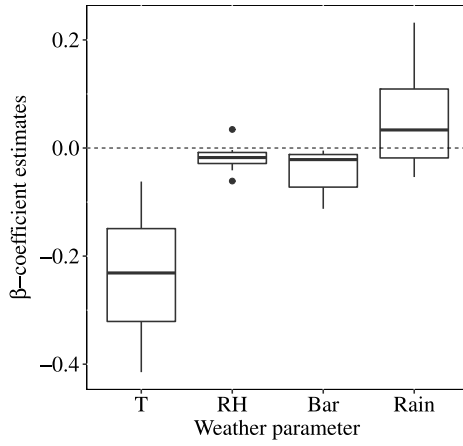


Fig. 3. Boxplots of the estimated β -coefficients for the weather parameters in the ARIMAX_{All} models.

computed one positive correlation (see the RSSI_05 in Table 3), all the β -coefficients are negative for T, indicating that cold temperatures improve the RSSI. This behavior is supported by the interpretation that higher temperatures generate more thermal noise, which will highly affect the quality of the received signal. It can be observed in different technologies [52] but LoRa is especially susceptible to this effect since it operates below the noise floor [32]. The RH and rain are other relevant conditions for the RSSI, having significant coefficients in five of the eight transmitter nodes. From Table 4 and Fig. 3, we note the RH coefficients are close to zero, having a negative direction for all nodes except RSSI_08. The rain is positive for all nodes with significant coefficients except RSSI_03. The Bar coefficients are negative in all the ARIMAX_{All} models (see the boxplot in Fig. 3), but have significant values only for two nodes (see Table 4). Therefore, although there are still discrepancies in the direction of the relationships, the results using the ARIMAX_{All} models are more coherent than those from the correlation analysis. As noted by [11], the discrepancies might be due to the distance between the LoRaWAN nodes and the gateway and the shadowing effects due to different obstructions in the communication channel path. The transmitted signal from a particular node has a different transmission time depending upon the distance from the gateway.

The rest of the analysis is devoted to fitting three other model configurations and comparing their forecasting results: (i) ARIMAX_{Main}, fitted only with the significant weather parameters; (ii) ARIMAX_{Temp}, fitted only with the temperature; (iii) ARIMA, without using the weather

parameters as regressors. All these models are fitted using the orders reported in Table 4.

Table 5 gives the values of all accuracy measures and the percentage difference when comparing these measures for the ARIMA with the other fitted models. These quantities are computed from the one-step-ahead forecasting using the testing set. It is worth to note that the inclusion of the weather parameters improved the accuracy measures for most of the nodes. The ARIMA was superior only for the RSSI_2 and RSSI_7 series, although with a difference smaller than 1%.

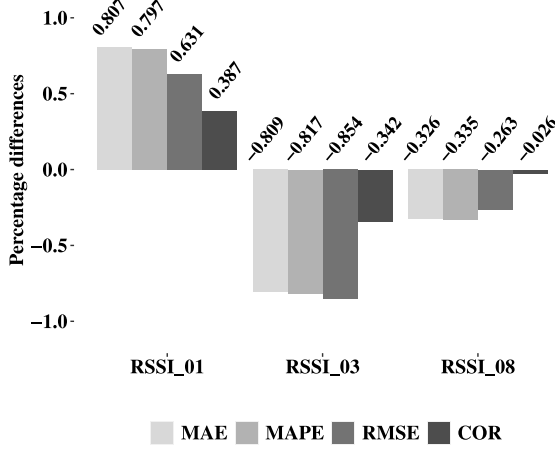
Table 6 counts the number of times each model yielded the best fit in each node, corroborating the importance of considering the weather parameters. The ARIMA without regressors outperforms the other models only for 12.5% (4/32) of the calculated measures. Despite of this, the difference in these metrics is really low, being smaller than 0.4% in all cases (see Table 5). The MAE of RSSI_02 is 1.474 for the ARIMA without regressors and 1.475 for the ARIMAX_{Temp}, while their MAPE is numerically equal. The other cases are the RMSE and COR of the RSSI_07. On the contrary, the accuracy measures show high differences most of the times that the alternatives with regressors outperform ARIMA, e.g., for the RSSI_06 the MAPE is 1.319 for the ARIMAX_{All} while for the ARIMA it is 1.450, resulting in a percentage difference of 9.069% in favor of the first (see Table 5). When comparing models with regressors, the ARIMAX_{All} and ARIMAX_{Temp} models are very competitive. The first presents the best results for 50% (16/32) of the calculated measures, while the ARIMAX_{Temp} outperforms the competitive models in 31.25% (10/32) of the cases. Moreover, these two configurations return similar values for all accuracy measures, e.g., for the RSSI_01 the percentage difference between their MAPEs is 0.804. For the RSSI_03, the ARIMAX_{All} provides worse results, e.g., using the ARIMAX_{All} provides a percentage difference of -0.81% when comparing its MAPE to the ARIMAX_{Temp}. These results are illustrated in Fig. 4, which presents this comparison for three selected LoRaWAN nodes. We highlight that ARIMAX_{Temp} is a parsimonious model, since it only requires the temperature instead of all the weather parameters, and provides consistently better fits than the ARIMA without regressors. Therefore, we selected this model as a benchmark in the remaining experiments and to be used in the first step of the hybrid methodology.

4.3. Fitting the pure AI algorithms

This section assesses the predictive capabilities of the pure AI algorithms. After applying Min-Max transformation for data preparation, we train the algorithms using $\rho = 8$ lagged values and the weather parameters as inputs. The pure ML algorithms considered are the ANN, RF, and SVM. They are fitted using the `neuralnet` [40], `randomForest` [45], and `svm` [43] R functions, respectively. The LSTM

Table 4Order for p , d , and q and significant regressors in the ARIMAX_{All} fitted model.

Node	Order			Weather condition ^a			
	p	d	q	T	RH	Bar	Rain
RSSI_01	2	1	3	negative			positive
RSSI_02	4	1	3	negative	negative	negative	positive
RSSI_03	4	1	2	negative	negative		negative
RSSI_04	3	1	3	negative		negative	
RSSI_05	2	1	3	negative			
RSSI_06	2	1	3	negative	negative		positive
RSSI_07	2	1	3	negative	negative		positive
RSSI_08	2	1	2	negative	positive		

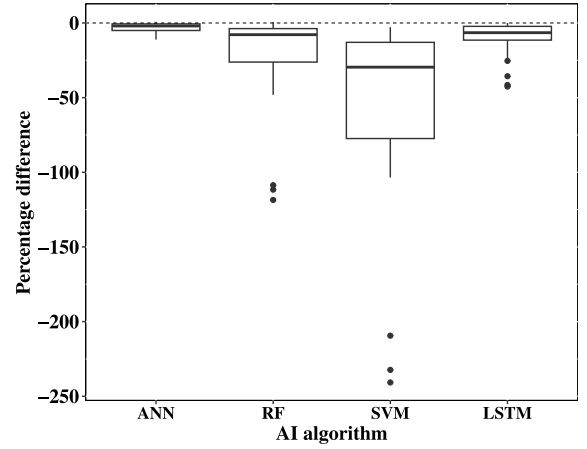
^a We set the significance level at 5%.**Fig. 4.** Percentage difference of the ARIMAX_{All} with respect to the ARIMAX_{Temp} at three LoRaWAN nodes.

network is fitted utilizing `ts.lstm` [53] function, which based on Keras and TensorFlow Python modules. Table 7 presents the accuracy measures and percentage differences of these algorithms compared to the ARIMAX_{Temp} model. The numerical results show that our benchmark model is the best option in 75% (24/32) of the computed accuracy measures (see Table 8).

The ANN is the best option among the pure AI algorithms considered. However, it still resulted in a negative mean percentage difference, suggesting that the ARIMAX_{Temp} model is more effective in computing the underlying trend of the RSSI signals than pure AI algorithms. The visual inspection of these differences, presented in Fig. 5, corroborates this conclusion. In addition, as noted by [11], even for the dataset of interest AI algorithms may require a long training time to implement the pre-training and fine-parameter tuning phases sequentially. On the contrary, the ARIMAX_{Temp} has the advantage of simplicity and does not require extensive feature engineering or complex steps. This can save time and energy on devices with limited computational resources during the implementation process, in line with the requirements of LoRaWAN and WSNs. Therefore, we use the ARIMAX_{Temp} as benchmark model to compare it with the hybrid methodology.

4.4. Fitting the hybrid models

Here, we evaluate the predictive capabilities of the hybrid models. The hybrid models are constructed using the best-fitted ARIMA model in the first stage, i.e., the ARIMAX_{Temp} models from Section 4.2. In the second stage we apply four different AI algorithms in the residual of the ARIMAX_{Temp}. The same algorithms fitted in the previous subsection are trained using $\rho = 4$ lagged residuals as inputs. Since the lagged residuals are already in the same scale as the residual outcome, it is not necessary

**Fig. 5.** Boxplots with the percentage difference of all accuracy measures for AI algorithms with respect to the ARIMAX_{Temp}.

to normalize them before training the AI models. Table 9 presents the accuracy measures and percentage differences of the hybrid models in comparison with those from the first stage. Table 10 gives the number of times each competitive model yielded the best forecasting performance. The numerical results show that the hybrid approach models outperform the ARIMAX_{Temp} models in 81.25% (26/32) of the computed accuracy measures (see Table 10). The ARIMA-ANN is the best option among the hybrid approaches. However, the percentage differences are very small. When taking only the positive differences, we note that they vary between 0.01% (MAPE of the ARIMA-ANN in RSSI_01) and 3.022% (MAPE of the ARIMA-SVM in RSSI_06), see Table 9 and Fig. 6. In addition, all hybrid models resulted in a negative mean percentage difference and from Fig. 7 we have visual inspection of these differences cumulatively. These results may suggest that the ARIMAX_{Temp} model can completely capture the underlying trend of the RSSI signals, which is the case when the variable of interest do not present nonlinear patterns.

Finally, Fig. 8 presents the ARIMAX_{Temp} and ARIMA-ANN forecasts versus the actual values in the last 100 h from the testing sets. This visual inspection indicates the ARIMAX_{Temp} and hybrid models have a similar performance and are quite accurately, corroborating the numerical results. The work [11] also found accurate forecasts when fitting a complex nonlinear ML algorithm to the same dataset [12] although with an additional pre-processing procedure and without presenting the accuracy measures of each RSSI test set. Moreover, we would like to mention that our results are not directly comparable to [11] since their study employs a distinct pre-processing process for the data, focuses solely on six of the LoRaWAN end devices, and only provides the global mean absolute error (MAE) for both test and validation datasets. Therefore, the accuracy measures of the RSSI test sets are not available in their study. Although the results are not comparable, making a visual inspection in Fig. 11 from [11] we note their algorithm has a similar performance to the proposed methodology. Hence, we can conclude that the ARIMAX_{Temp} and ARIMA-ANN are the two most competitive methods, with the former having advantage since it provided accuracy measures slightly better in the current experiment. Moreover, differently from LSTM-based methods [11], the ARIMA methodology has the advantage of being simpler to be implemented in devices and allowing to compute the direction of the relationship between the RSSI and a weather condition of interest. We selected these two most competitive models, i.e., the ARIMAX_{Temp} and ARIMA-ANN, to perform the remaining experiments.

Table 5

Accuracy measures of each ARIMA configuration for one-step-ahead forecasting in the testing sets and percentage difference with respect to the configuration without regressors. ARIMAX_{All} refers to the complete model considering all the weather parameters as regressors, ARIMAX_{Main} refers to the model fitted only with the significant weather parameters, ARIMAX_{Temp} refers to the model fitted only with the temperature, and ARIMA to the model without using the weather parameters as regressors.

Node	Measure	Model				Percentage difference		
		ARIMAX _{All}	ARIMAX _{Main}	ARIMAX _{Temp}	ARIMA	ARIMAX _{All}	ARIMAX _{Main}	ARIMAX _{Temp}
RSSI_01	MAE	0.510	0.513	0.514	0.542	5.975	5.455	5.210
	MAPE	0.914	0.919	0.921	0.972	5.997	5.486	5.242
	RMSE	0.661	0.665	0.665	0.701	5.62	5.12	5.021
	COR	0.784	0.782	0.781	0.754	4.081	3.742	3.680
RSSI_02	MAE	1.476	1.476	1.475	1.474	-0.089	-0.089	-0.055
	MAPE	1.793	1.793	1.790	1.790	-0.124	-0.124	-0.001
	RMSE	2.034	2.034	2.034	2.043	0.448	0.448	0.418
	COR	0.854	0.854	0.854	0.852	0.223	0.223	0.205
RSSI_03	MAE	0.483	0.48	0.479	0.484	0.279	0.861	1.079
	MAPE	0.721	0.717	0.715	0.723	0.287	0.876	1.095
	RMSE	0.607	0.603	0.602	0.612	0.736	1.332	1.576
	COR	0.836	0.838	0.839	0.833	0.351	0.592	0.696
RSSI_04	MAE	0.557	0.562	0.554	0.578	3.542	2.799	4.042
	MAPE	0.721	0.726	0.717	0.748	3.568	2.83	4.079
	RMSE	0.689	0.695	0.695	0.719	4.21	3.3	3.299
	COR	0.902	0.9	0.899	0.891	1.203	0.984	0.893
RSSI_05	MAE	0.897	0.903	0.903	0.908	1.197	0.544	0.544
	MAPE	1.217	1.225	1.225	1.233	1.339	0.668	0.668
	RMSE	1.18	1.19	1.19	1.212	2.693	1.85	1.850
	COR	0.945	0.944	0.944	0.941	0.342	0.236	0.236
RSSI_06	MAE	0.786	0.79	0.807	0.862	8.745	8.392	6.323
	MAPE	1.319	1.323	1.353	1.45	9.069	8.741	6.685
	RMSE	1.135	1.14	1.16	1.249	9.09	8.684	7.125
	COR	0.935	0.934	0.932	0.92	1.56	1.483	1.228
RSSI_07	MAE	0.406	0.405	0.406	0.405	-0.088	0.013	-0.213
	MAPE	0.661	0.66	0.661	0.66	-0.081	0.024	-0.190
	RMSE	0.538	0.538	0.54	0.536	-0.404	-0.377	-0.812
	COR	0.747	0.747	0.745	0.75	-0.317	-0.296	-0.645
RSSI_08	MAE	0.828	0.829	0.825	0.838	1.248	1.153	1.569
	MAPE	0.968	0.969	0.965	0.981	1.28	1.191	1.609
	RMSE	1.078	1.08	1.076	1.084	0.476	0.339	0.737
	COR	0.961	0.961	0.961	0.961	0.016	0.004	0.042
Mean $\pm$ SD						2.265 $\pm$ 2.893	2.078 $\pm$ 2.696	1.976 $\pm$ 2.318

Table 6

Times that each ARIMA configuration presented the best accuracy in each node.

Node	ARIMAX	ARIMAX _{Main}	ARIMAX _{Temp}	ARIMA
RSSI_01	4	0	0	0
RSSI_02	2 ^a	0	0	2
RSSI_03	0	0	4	0
RSSI_04	2	0	2	0
RSSI_05	4	0	0	0
RSSI_06	4	0	0	0
RSSI_07	0	2	0	2
RSSI_08	0	0	4	0
Overall	16	2	10	4

^a The ARIMAX_{All} and ARIMAX_{Main} are equal.

4.5. Missing values experiment

Missing values are ubiquitous in data from received LoRa signals, especially in outdoor environments, which can greatly degrade the localization accuracy [54]. One solution for this situation is to perform data imputation, which is based on the idea of substituting the missing values with an estimate of the distribution of this variable [55]. The simplest way to perform data imputation is using the mean value of all of the data to interpolate the missing values. However, it cannot reflect the temporal changes that are inherent to univariate time series. Another alternative is the MA imputation method, which interpolates the missing value with the mean value of the moving window. This method may perform well when the percentage of missing values is low and the missingness occurs randomly. However, LoRa signals can

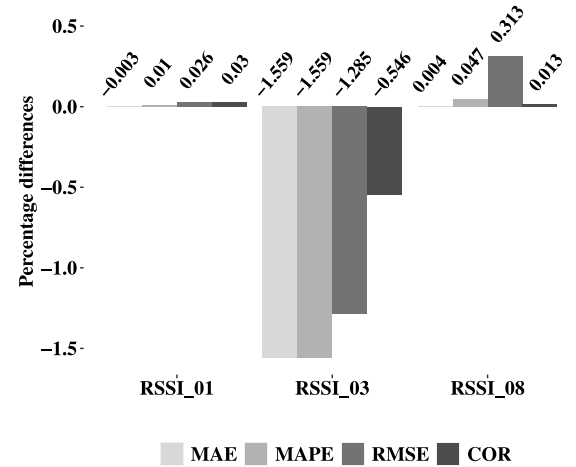


Fig. 6. Percentage difference of the ARIMA-ANN with respect to the ARIMAX_{Temp} at three LoRaWaN nodes.

be sometimes missing for a large period due to failures of the devices, generating a gap in the time series. In this context, since the proposed methodologies take the time effect in the forecasting procedure, they may represent a feasible alternative to perform data imputation to replace RSSI missing values. Therefore, in this section, we perform an experiment to evaluate this simulating different gap sizes and the

Table 7Accuracy measures of the pure ML algorithms for one-step-ahead forecasting in the testing sets and percentage difference with respect to the ARIMAX_{Temp}.

Node	Measure	Model					Percentage difference			
		ANN	RF	SVM	LSTM	ARIMAX _{Temp}	ANN	RF	SVM	LSTM
RSSI_01	MAE	0.538	0.546	0.587	0.546	0.514	-4.743	-6.183	-14.216	-6.318
	MAPE	0.965	0.979	1.053	0.977	0.921	-4.710	-6.236	-14.296	-6.072
	RMSE	0.692	0.693	0.739	0.710	0.665	-4.012	-4.209	-11.013	-6.661
	COR	0.761	0.763	0.721	0.760	0.781	-2.676	-2.401	-7.737	-2.743
RSSI_02	MAE	1.575	1.922	2.671	1.625	1.475	-6.780	-30.261	-81.039	-10.171
	MAPE	1.924	2.347	3.317	1.991	1.790	-7.434	-31.089	-85.270	-11.178
	RMSE	2.153	2.539	3.584	2.268	2.034	-5.863	-24.816	-76.183	-11.478
	COR	0.845	0.810	0.663	0.835	0.854	-1.004	-5.137	-22.403	-2.164
RSSI_03	MAE	0.488	0.477	0.521	0.485	0.479	-1.829	0.551	-8.817	-1.171
	MAPE	0.729	0.711	0.779	0.724	0.715	-1.960	0.557	-8.968	-1.314
	RMSE	0.615	0.603	0.655	0.618	0.602	-2.176	-0.218	-8.845	-2.732
	COR	0.836	0.843	0.814	0.836	0.839	-0.326	0.402	-3.052	-0.372
RSSI_04	MAE	0.565	0.595	0.697	0.618	0.554	-1.983	-7.325	-25.794	-11.411
	MAPE	0.732	0.770	0.902	0.799	0.717	-2.037	-7.387	-25.718	-11.482
	RMSE	0.709	0.753	0.932	0.772	0.695	-1.990	-8.264	-34.074	-11.105
	COR	0.895	0.881	0.811	0.878	0.899	-0.456	-1.969	-9.816	-2.280
RSSI_05	MAE	0.896	1.020	1.191	0.916	0.903	0.780	-12.937	-31.837	-1.439
	MAPE	1.219	1.381	1.617	1.243	1.225	0.487	-12.777	-32.043	-1.509
	RMSE	1.198	1.348	1.591	1.221	1.190	-0.648	-13.322	-33.677	-2.644
	COR	0.943	0.936	0.917	0.944	0.944	-0.054	-0.767	-2.784	0.024
RSSI_06	MAE	0.895	1.196	1.643	1.012	0.807	-10.880	-48.127	-103.487	-25.381
	MAPE	1.503	1.991	2.689	1.695	1.353	-11.063	-47.136	-98.690	-25.220
	RMSE	1.243	1.518	2.310	1.380	1.160	-7.132	-30.834	-99.149	-19.011
	COR	0.924	0.919	0.805	0.910	0.932	-0.841	-1.342	-13.606	-2.361
RSSI_07	MAE	0.412	0.446	0.556	0.453	0.406	-1.443	-9.907	-36.950	-11.583
	MAPE	0.670	0.726	0.902	0.736	0.661	-1.370	-9.768	-36.353	-11.364
	RMSE	0.543	0.579	0.725	0.589	0.540	-0.488	-7.101	-34.134	-9.094
	COR	0.746	0.713	0.541	0.705	0.745	0.099	-4.250	-27.335	-5.295
RSSI_08	MAE	0.919	1.806	2.876	1.224	0.865	-6.213	-108.633	-232.347	-41.484
	MAPE	1.076	2.142	3.450	1.443	1.012	-6.277	-111.622	-240.760	-42.568
	RMSE	1.153	2.428	3.438	1.507	1.111	-3.816	-118.525	-209.417	-35.656
	COR	0.962	0.849	0.794	0.951	0.959	0.292	-11.455	-17.154	-0.787
Mean $\pm$ SD							-3.08 $\pm$ 3.19	-21.33 $\pm$ 32.60	-52.72 $\pm$ 64.50	-10.44 $\pm$ 11.70

Table 8Times that each pure ML algorithms presented the best accuracy in comparison with the ARIMAX_{Temp} in each node.

Node	ANN	RF	SVM	LSTM	ARIMAX _{Temp}
RSSI_01	0	0	0	0	4
RSSI_02	0	0	0	0	4
RSSI_03	0	3	0	0	1
RSSI_04	0	0	0	0	4
RSSI_05	2	0	0	1	1
RSSI_06	0	0	0	0	4
RSSI_07	1	0	0	0	3
RSSI_08	1	0	0	0	3
Overall	4	3	0	1	24

advantage of using the proposed ARIMA approaches instead of the standard MA imputation method.

The experiment is carried out by supposing the testing sets have missing values in different parts of the series. For each scenario, the missing values are imputed using the ARIMAX_{Temp} and ARIMA-ANN, which represent the best alternatives regarding the ARIMA and hybrid approaches, respectively. The experiment procedure is given by the following algorithm:

1. Let n be the sample size in the training set and y_i , $i = n+1, \dots, n+374$, the i th observation of the dataset, which is part of the test set.
2. For each y_i , suppose that $Y = \{y_i, y_{i+1}, \dots, y_{i+k-1}\}$ is a sequence of missing values with gap size k .
3. Use the ARIMAX_{Temp} and ARIMA-ANN models previously trained to forecast Y based on the information up to y_{i-1} and compute the MA imputation with a moving window of size k .

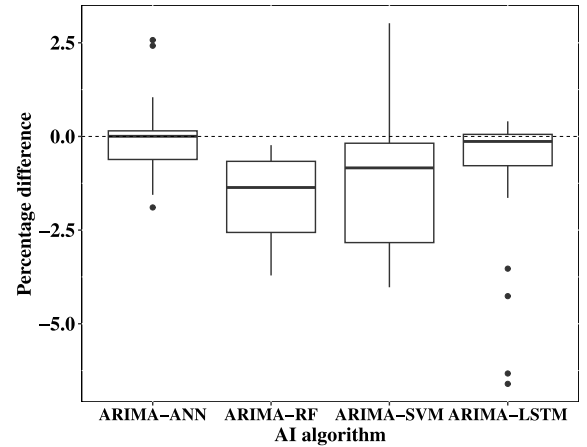


Fig. 7. Boxplots with the percentage difference of all accuracy measures for the hybrid algorithms with respect to the ARIMAX_{Temp}.

4. Compute the RMSE of each imputation method varying $i \in 1, \dots, 374 - k + 1$ and, at the end of these calculations, compute the average of the RMSE of each imputation method.
5. Steps 1 to 4 are repeated for $k = 4$, $k = 19$, and $k = 37$, which represent gap sizes of 1%, 5%, and 10% of the testing sets, respectively.

The average RMSEs of each imputation method for LoRaWAN node are shown in Fig. 9. As expected, the accuracy measures are higher as the gap sizes increase. Generally, the ARIMAX_{Temp} and ARIMA-ANN

Table 9Accuracy measures of the hybrid algorithms for one-step-ahead forecasting in the testing sets and percentage difference with respect to the ARIMAX_{Temp}.

Node	Measure	Model					Percentage difference			
		ARIMA-ANN	ARIMA-RF	ARIMA-SVM	ARIMA-LSTM	ARIMAX _{Temp}	ARIMA-ANN	ARIMA-RF	ARIMA-SVM	ARIMA-LSTM
RSSI_01	MAE	0.514	0.522	0.531	0.514	0.514	-0.003	-1.614	-3.389	0.016
	MAPE	0.921	0.936	0.953	0.920	0.921	0.010	-1.618	-3.399	0.142
	RMSE	0.665	0.681	0.692	0.671	0.665	0.026	-2.349	-4.026	-0.858
	COR	0.782	0.771	0.764	0.782	0.781	0.030	-1.364	-2.192	0.050
RSSI_02	MAE	1.474	1.495	1.498	1.481	1.475	0.050	-1.349	-1.532	-0.422
	MAPE	1.791	1.815	1.819	1.799	1.790	-0.020	-1.350	-1.620	-0.476
	RMSE	2.026	2.047	2.047	2.053	2.034	0.391	-0.609	-0.609	-0.921
	COR	0.855	0.851	0.852	0.853	0.854	0.145	-0.360	-0.183	-0.104
RSSI_03	MAE	0.487	0.486	0.477	0.478	0.479	-1.559	-1.444	0.437	0.208
	MAPE	0.726	0.725	0.712	0.713	0.715	-1.559	-1.406	0.489	0.231
	RMSE	0.610	0.609	0.606	0.600	0.602	-1.285	-1.124	-0.612	0.275
	COR	0.835	0.835	0.837	0.840	0.839	-0.546	-0.489	-0.266	0.062
RSSI_04	MAE	0.561	0.569	0.555	0.589	0.554	-1.115	-2.558	-0.177	-6.328
	MAPE	0.725	0.736	0.718	0.765	0.717	-1.096	-2.576	-0.174	-6.604
	RMSE	0.708	0.713	0.700	0.725	0.695	-1.896	-2.629	-0.648	-4.263
	COR	0.895	0.893	0.897	0.901	0.899	-0.465	-0.683	-0.165	0.211
RSSI_05	MAE	0.911	0.933	0.933	0.900	0.903	-0.835	-3.240	-3.287	0.404
	MAPE	1.235	1.262	1.261	1.221	1.225	-0.822	-3.030	-2.942	0.305
	RMSE	1.196	1.234	1.228	1.189	1.190	-0.497	-3.689	-3.168	0.054
	COR	0.943	0.940	0.940	0.944	0.944	-0.054	-0.417	-0.350	0.023
RSSI_06	MAE	0.788	0.812	0.783	0.819	0.807	2.421	-0.588	3.021	-1.388
	MAPE	1.318	1.360	1.312	1.375	1.353	2.573	-0.491	3.022	-1.641
	RMSE	1.148	1.178	1.168	1.201	1.160	1.046	-1.554	-0.722	-3.530
	COR	0.933	0.929	0.931	0.931	0.932	0.160	-0.233	-0.062	-0.070
RSSI_07	MAE	0.406	0.411	0.410	0.407	0.406	0.099	-1.296	-0.958	-0.148
	MAPE	0.661	0.670	0.668	0.663	0.661	0.078	-1.362	-1.047	-0.199
	RMSE	0.541	0.548	0.549	0.543	0.540	-0.071	-1.355	-1.604	-0.494
	COR	0.745	0.738	0.736	0.744	0.745	-0.028	-0.971	-1.138	-0.119
RSSI_08	MAE	0.826	0.857	0.860	0.834	0.832	0.728	-3.070	-3.457	-0.262
	MAPE	0.965	1.002	1.006	0.975	0.972	0.759	-3.054	-3.462	-0.213
	RMSE	1.074	1.121	1.111	1.089	1.081	0.668	-3.711	-2.798	-0.756
	COR	0.961	0.958	0.958	0.960	0.961	0.043	-0.334	-0.257	-0.074
Mean $\pm$ SD							-0.082 $\pm$ 0.981	-1.622 $\pm$ 1.049	-1.165 $\pm$ 1.737	-0.84 $\pm$ 1.795

Table 10Times that each hybrid algorithm presented the best accuracy in comparison with the ARIMAX_{Temp} in each node.

Node	ARIMA-ANN	ARIMA-RF	ARIMA-SVM	ARIMA-LSTM	ARIMAX _{Temp}
RSSI_01	1	0	0	3	0
RSSI_02	3	0	0	0	1
RSSI_03	0	0	2	2	0
RSSI_04	0	0	0	1	3
RSSI_05	0	0	0	4	0
RSSI_06	2	0	2	0	0
RSSI_07	2	0	0	0	2
RSSI_08	4	0	0	0	0
Overall	12	0	4	10	6

models are more reliable than the MA imputation method since they provide lower average RMSEs. Thus, both fitted models can improve the imputation performance and reduce the adverse effects caused by the missing data. When comparing the ARIMAX_{Temp} and ARIMA-ANN algorithms, we observe similar behaviors to those from the forecasting analysis in Table 10, i.e., these two configurations return very close values, yielding measures numerically equal for several scenarios. We note the higher differences between the ARIMAX_{Temp} and ARIMA-ANN algorithms is 0.136 for the RSSI_06 with a gap size of 10%. In this scenario, the ARIMAX_{Temp} provides the smaller RMSE, with an improvement of 0.671 with respect to the MA imputation method. When taking the differences between the ARIMAX_{Temp} and ARIMA-ANN algorithms, we note the higher value is 0.136 for the RSSI_06 and the gap size of 10%. These results corroborate the ability of the ARIMAX_{Temp} to capture the variations of the RSSI signals over time, being a useful model not only to forecast the RSSI values but also as an data imputation tool for dealing with missing values. Therefore, considering both methodologies

have a similar behavior, the ARIMAX_{Temp} model is preferred over the ARIMA-ANN because of its simplicity and interpretability.

5. Conclusion

Fluctuations in weather parameters can impact the reliability and connectivity of LoRaWAN networks. This effect is particularly relevant for outdoor WSNs, which are more exposed to environmental changes in temperature, barometric pressure, humidity, and rain. This paper investigated the effects of these referred weather parameters on the performance of LoRaWAN networks. We also evaluated if those variables can contribute to forecasting the RSSI measurements and their usefulness in performing data imputation in sequences with missing values. The results are obtained using the ARIMA and AI techniques including the weather parameters as regressors and coupling them through hybrid approaches. Therefore, we provided a comparative study of ARIMA and hybrid approaches to forecasting the RSSI considering variations on the temperature, pressure, relative humidity, and rain.

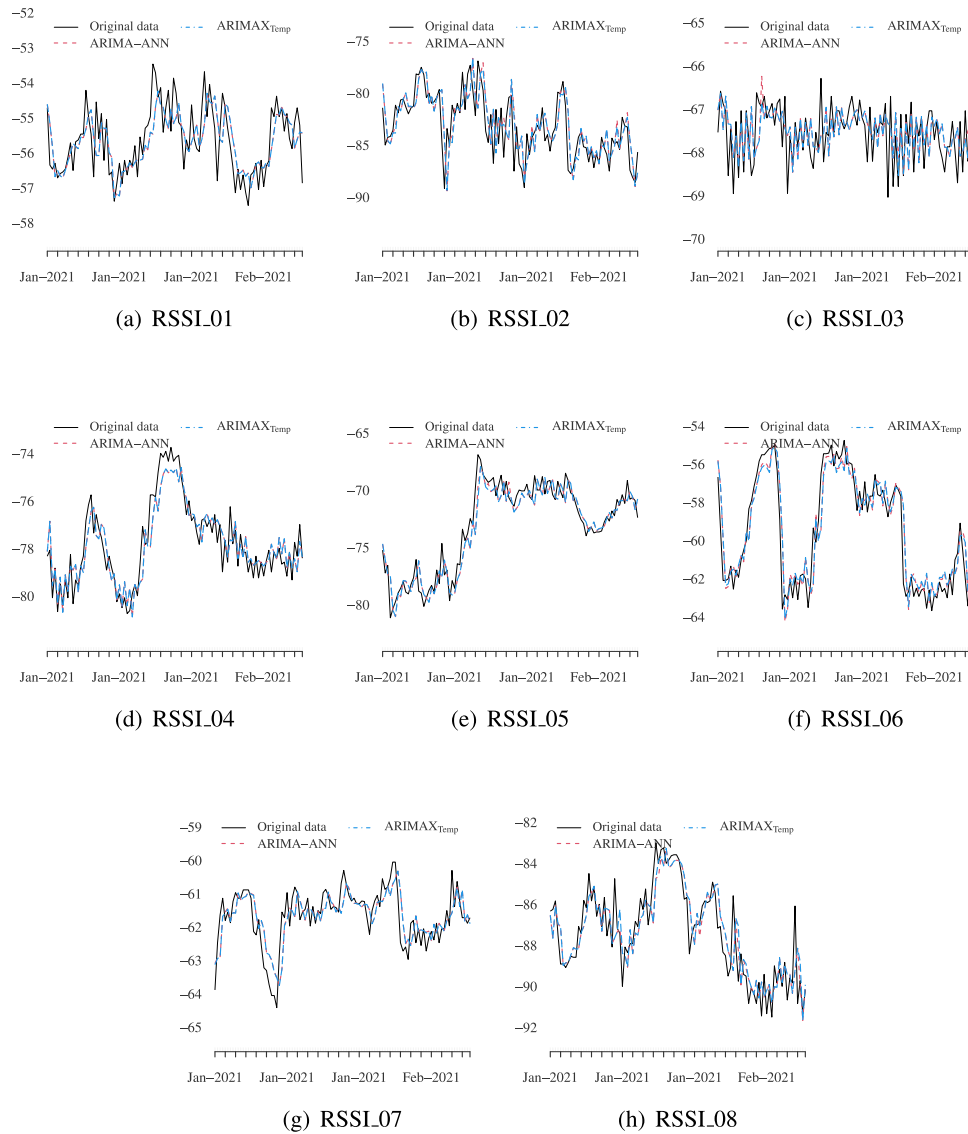


Fig. 8. Actual RSSI values versus the ARIMAX_{Temp} and the ARIMA-ANN forecasts in the last 100 h from the testing sets of each LoRaWAN node.

Our analysis confirmed the importance of considering weather parameters and using time series models to accurately analyze and forecast RSSI measurements. The temperature negatively impacted RSSI and was the most influential factor among the examined weather parameters. It is worth noting that the ARIMAX_{Temp} outperformed all the pure ML and LSTM algorithms and presented an equivalent performance with the hybrid approach that combined the ARIMA model with the temperature and the ANN. This is an interesting result to illustrate that, despite the undeniable relevance of AI algorithms, ARIMA models can be interesting competitors in some specific situations. Regarding the evaluation of the proposed methodologies to perform data imputation, they shown more reliability than the MA imputation method, providing lower average RMSEs and reducing the adverse effects caused by the missing data. Nevertheless, the accuracy measures obtained from this hybrid model closely resembled those achieved by the ARIMA model with the temperature regressor. Considering the similar performance of both methodologies, we recommend using the ARIMA model with the temperature over the ARIMA-ANN approach due to its simplicity and interpretability. These findings indicate that the RSSI of the analyzed LoRaWAN transmitter nodes is influenced by outdoor temperature.

CRediT authorship contribution statement

Renata Rojas Guerra: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Anna Vizziello:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Pietro Savazzi:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Emanuele Goldoni:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Paolo Gamba:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to the data and to my code in the manuscript.

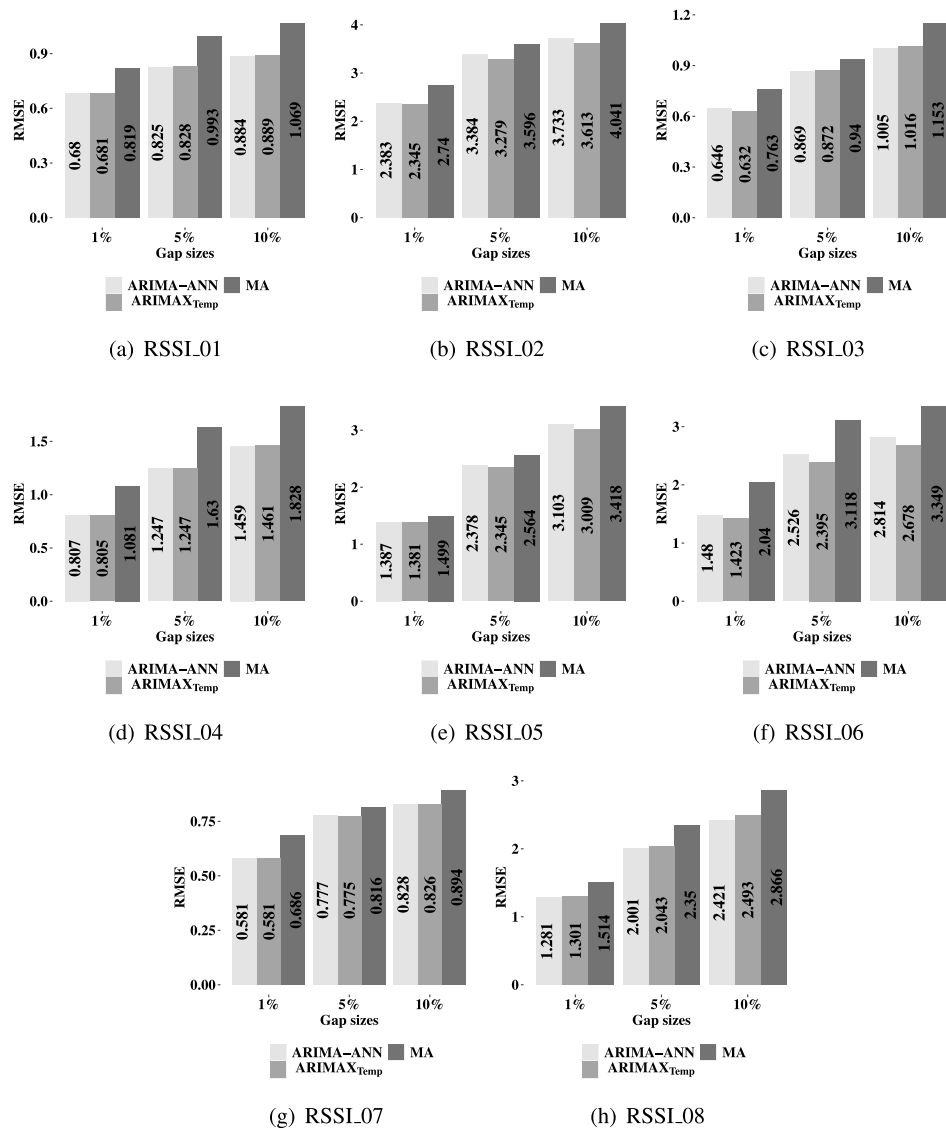


Fig. 9. The average RMSE of three imputation methods for the RSSI at the eight LoRaWAN nodes under 1% 5% and 10% gap sizes in the testing sets.

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